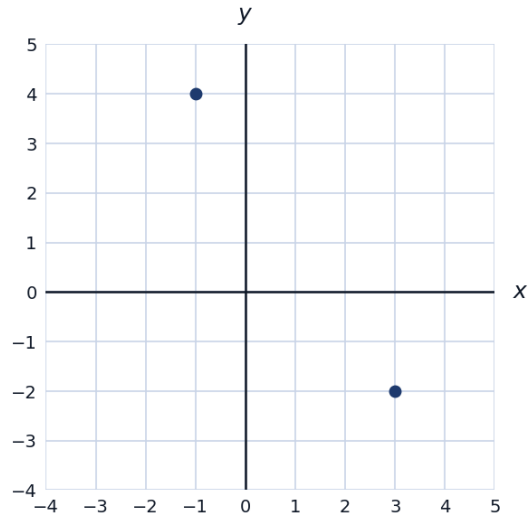


# Vectors: Components and Operations



## Vector arithmetic

- 1 Let  $\mathbf{u} = \langle 3, -2 \rangle$  and  $\mathbf{v} = \langle -1, 4 \rangle$ . Find:

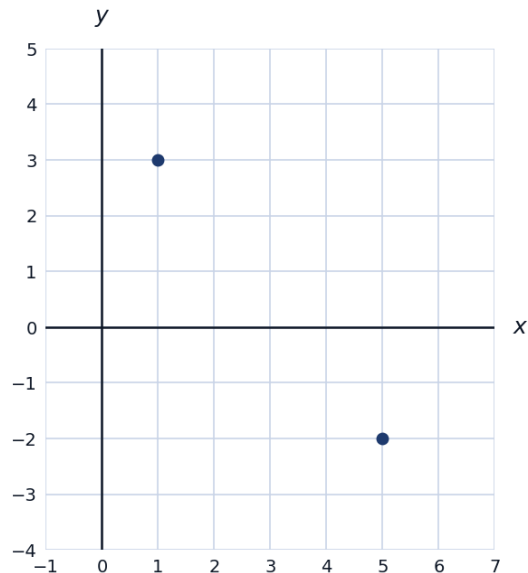


a  $\mathbf{u} + \mathbf{v}$

b  $3\mathbf{u} - 2\mathbf{v}$

c  $|\mathbf{u}|$

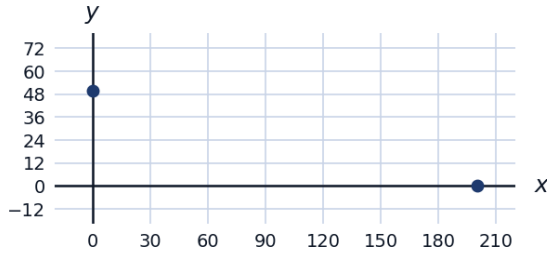
- 2 Find the component form and magnitude of the vector from  $P(1, 3)$  to  $Q(5, -2)$ .



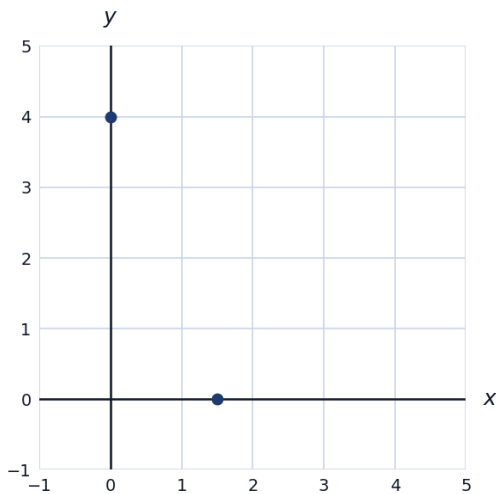
- 3 Find the unit vector in the direction of  $\mathbf{v} = \langle 6, -8 \rangle$ .
- 4 Write the vector of magnitude 12 making an angle of  $150^\circ$  with the positive x-axis in component form (exact values).

## Applications

- 5 A plane flies due east at 200 mph while the wind blows due north at 50 mph.



- a Find the magnitude of the plane's resultant velocity, to one decimal place.
- b Find the direction of the resultant, as an angle north of east to two decimal places.
- 6 Two forces  $\mathbf{F}_1 = \langle 30, 0 \rangle$  N and  $\mathbf{F}_2 = \langle 0, 40 \rangle$  N act on an object. Find the magnitude of the resultant force.
- 7 A boat aims to cross a river by heading due north at 4 m/s, but the current pushes it east at 1.5 m/s.

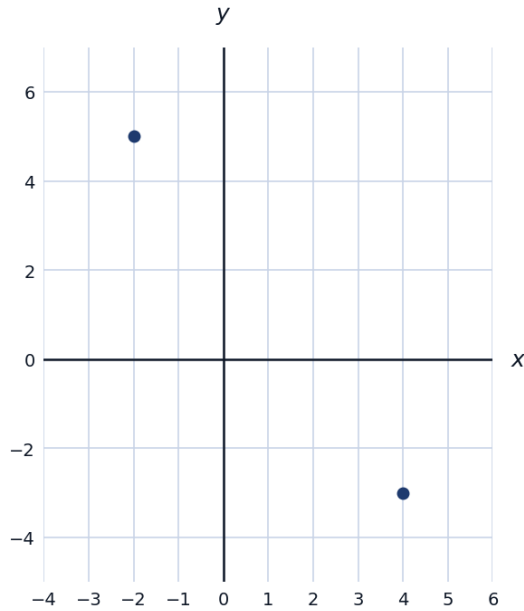


- a Find the boat's resultant speed, to two decimal places.
- b If the river is 200 m wide, find how long the crossing takes.
- c Find how far downstream (east) the boat lands.

---

## Position vectors and geometry

- 8 Given  $A(-2, 5)$  and  $B(4, -3)$ , find the midpoint of  $\overline{AB}$  and the vector  $\vec{AB}$ .

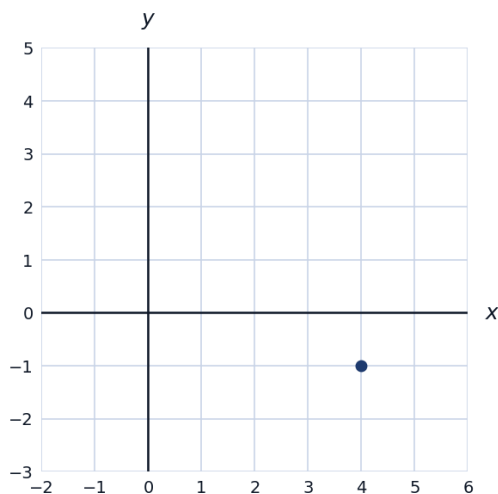


- 9 Let  $\mathbf{a} = \langle 2, 3 \rangle$  and  $\mathbf{b} = \langle -4, 1 \rangle$ . Find a vector  $\mathbf{c}$  such that  $\mathbf{a} + \mathbf{b} + \mathbf{c} = \langle 0, 0 \rangle$ , and verify your answer.
- 10 Express  $\mathbf{v} = \langle 5, -12 \rangle$  in terms of the standard unit vectors  $\mathbf{i}$  and  $\mathbf{j}$ , and find  $|\mathbf{v}|$ .

---

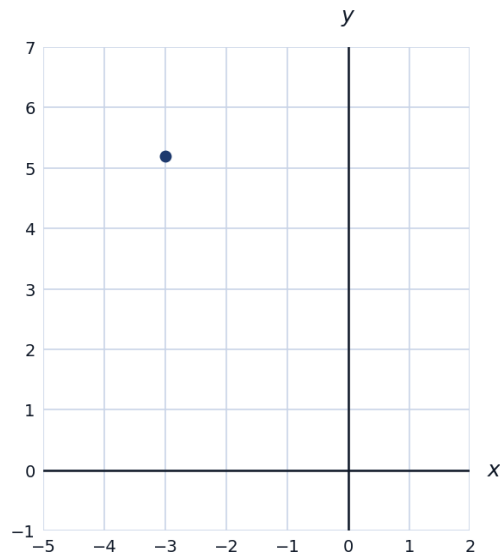
### More applications

- 11 An airplane has an airspeed of 305 mph and flies on a heading  $7.5^\circ$  west of due north, while a 40 mph wind blows due east. Find the plane's resultant ground velocity vector, and its ground speed, to one decimal place (take east as the positive x-direction and north as the positive y-direction).
- 12 Given  $\mathbf{u} = \langle 4, -1 \rangle$ , find a vector orthogonal to  $\mathbf{u}$  with the same magnitude as  $\mathbf{u}$ .



## Finding direction angles

- 13 Find the direction angle of  $\mathbf{v} = \langle -3, 3\sqrt{3} \rangle$ , to the nearest degree.



- 14 A vector has magnitude 10 and direction angle  $200^\circ$ . Write it in component form, rounded to two decimal places.

---

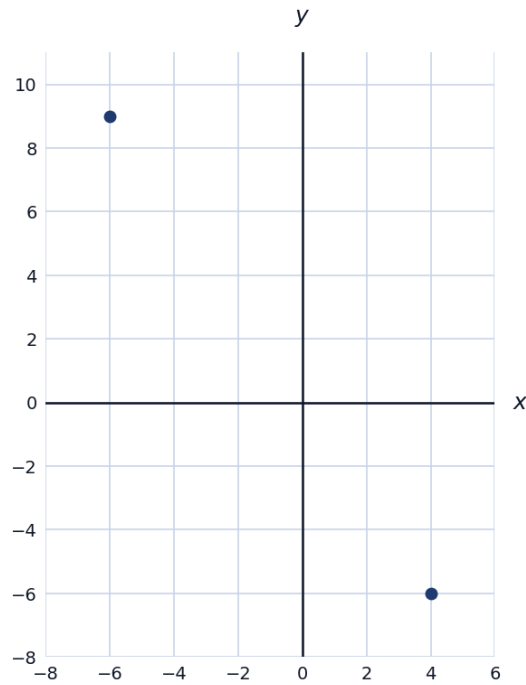
## Vector equations and unknowns

- 15 Find scalars  $c_1$  and  $c_2$  such that  $c_1\langle 1, 0 \rangle + c_2\langle 0, 1 \rangle = \langle -3, 7 \rangle$ .
- 16 Find scalars  $c_1$  and  $c_2$  such that  $c_1\langle 2, 1 \rangle + c_2\langle 1, 3 \rangle = \langle 7, 11 \rangle$ .

---

## Parallel and equal vectors

- 17 Determine whether  $\mathbf{u} = \langle 4, -6 \rangle$  and  $\mathbf{v} = \langle -6, 9 \rangle$  are parallel, and if so, state whether they point in the same or opposite directions.

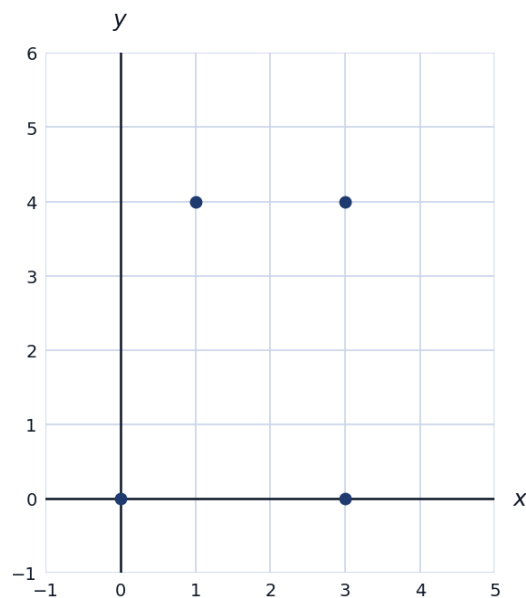


- 18 Find the value of  $k$  so that  $\langle k, 6 \rangle$  is parallel to  $\langle 4, 8 \rangle$ .

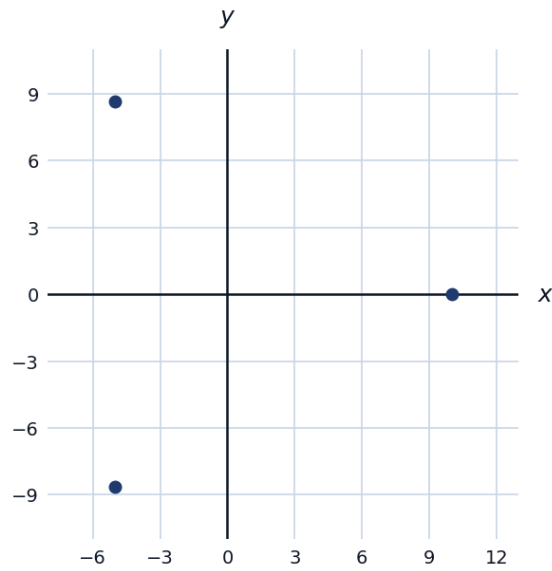
---

### Vectors in three real-world scenarios

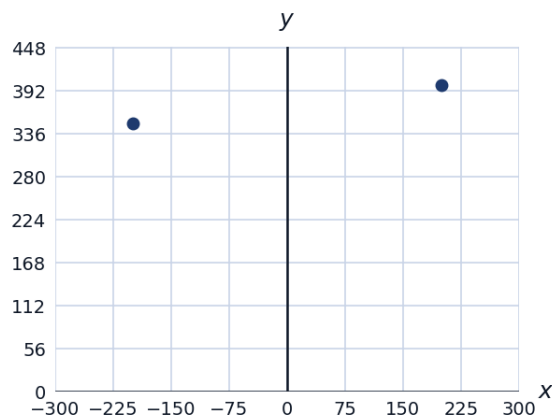
- 19 A hiker walks 3 km east, then 4 km north, then 2 km west. Find the magnitude of the hiker's total displacement from the starting point.



- 20 Three ropes pull on a ring with forces  $\langle 10, 0 \rangle$ ,  $\langle -5, 8.66 \rangle$ , and  $\langle -5, -8.66 \rangle$  N. Find the net force, and explain what this implies physically.



- 21 A crane hoists a load using two cables. One cable exerts a force of  $\langle -200, 350 \rangle$  N and the other exerts  $\langle 200, 400 \rangle$  N. Find the combined lifting force (the vertical component of the resultant), and explain why the horizontal components matter for the crane's structural design even though they cancel.



- 22** A swimmer aims to swim directly across a river at  $1.5 \text{ m/s}$  relative to the water, while the current flows at  $0.8 \text{ m/s}$  downstream. Find the swimmer's resultant speed relative to the shore, and the angle (from the intended straight-across direction) at which they actually drift, to the nearest degree.

